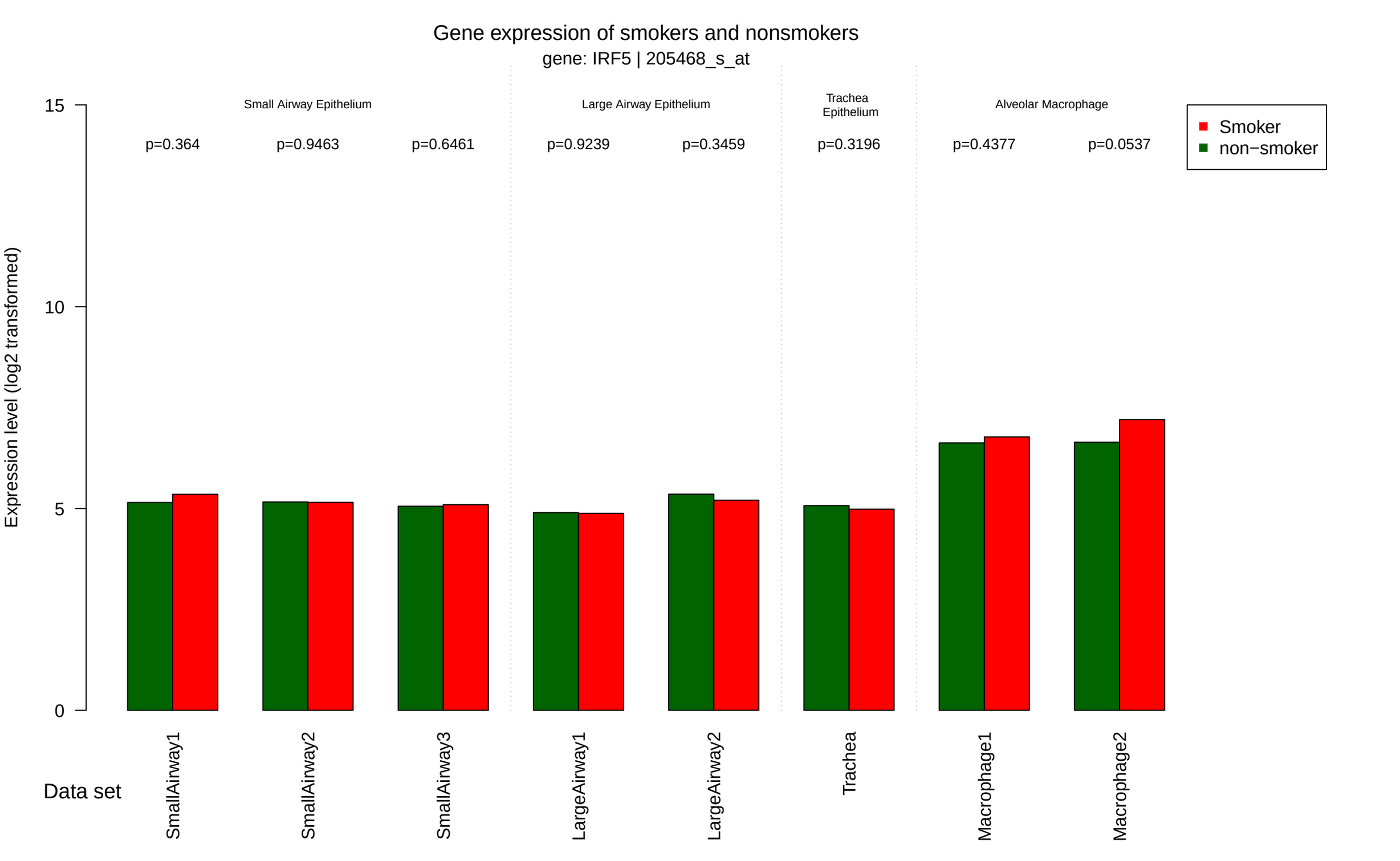
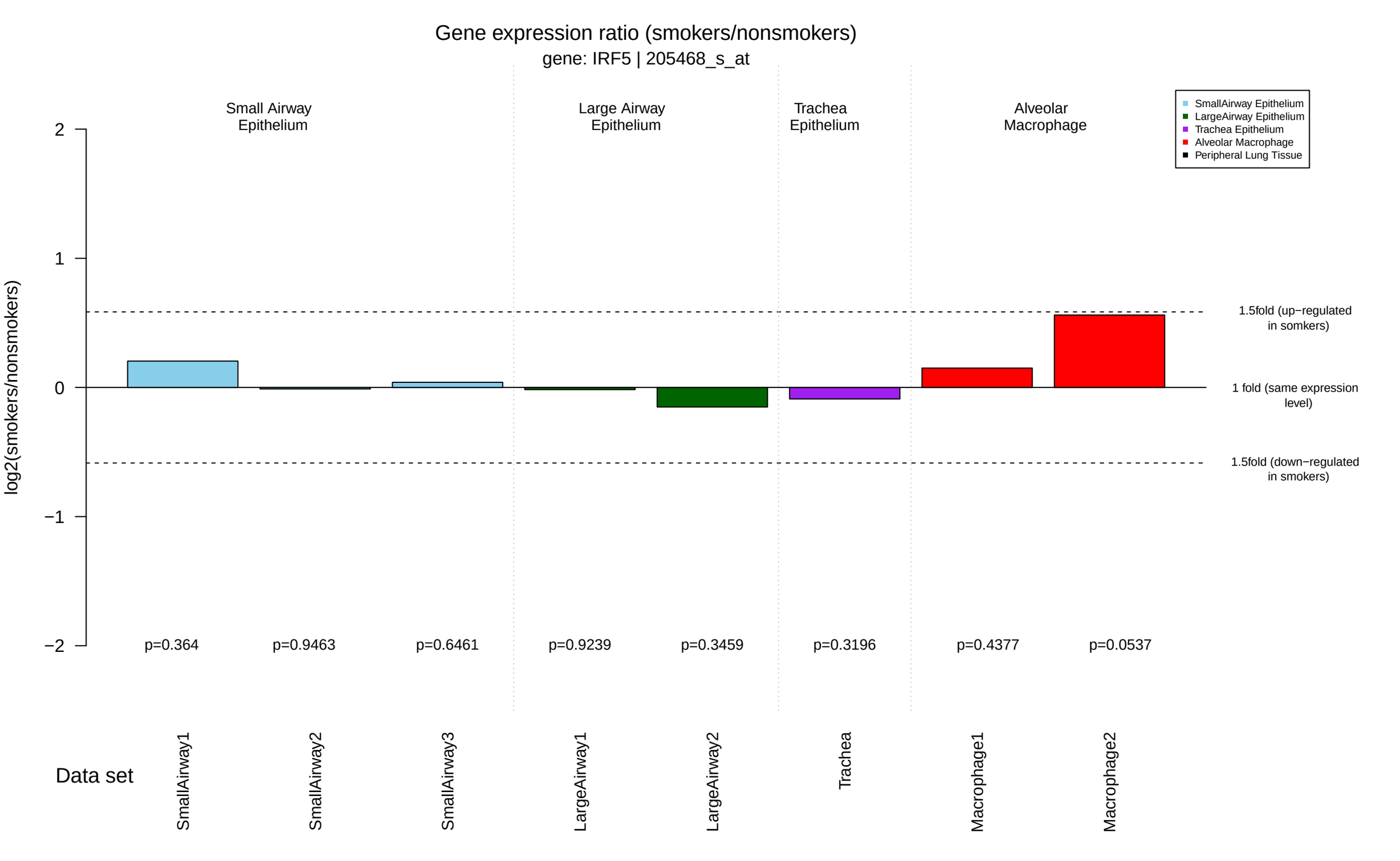
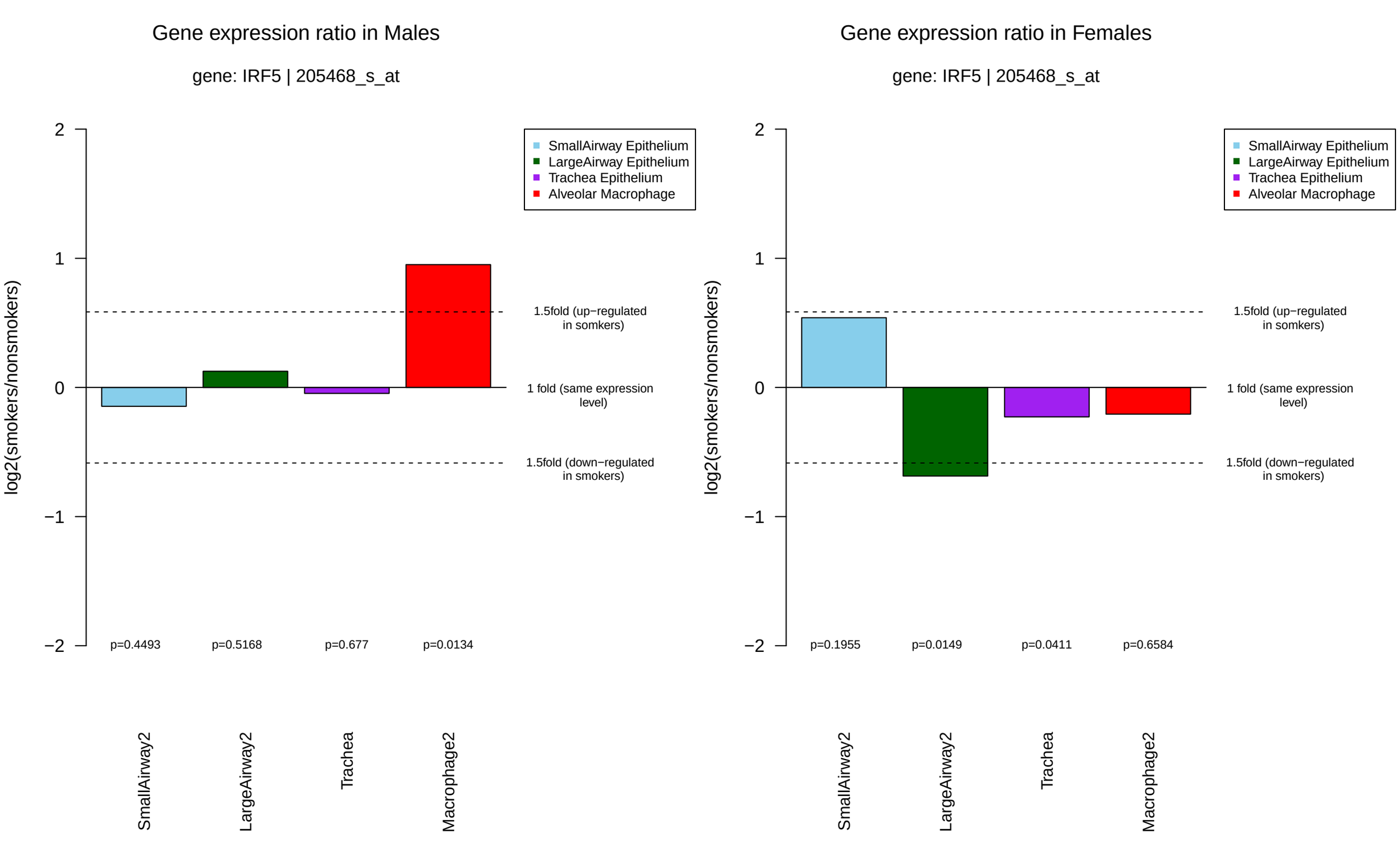
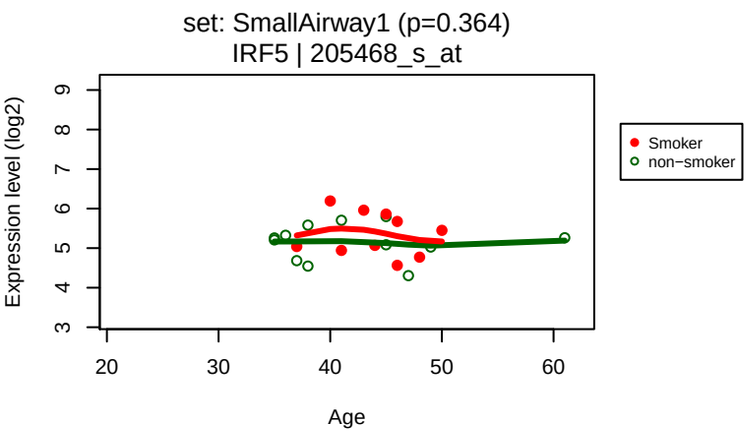
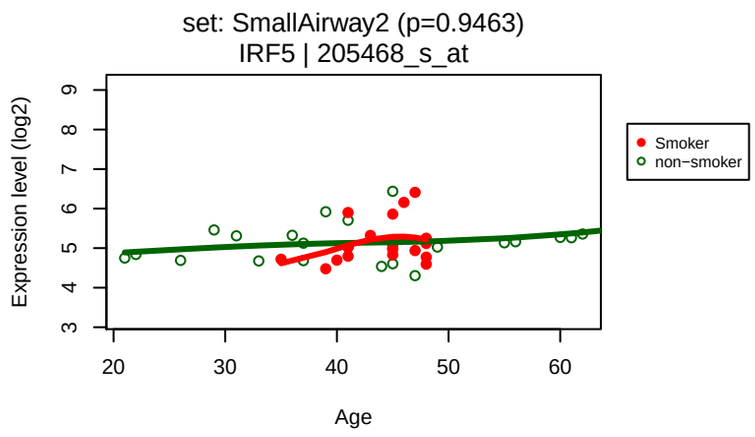
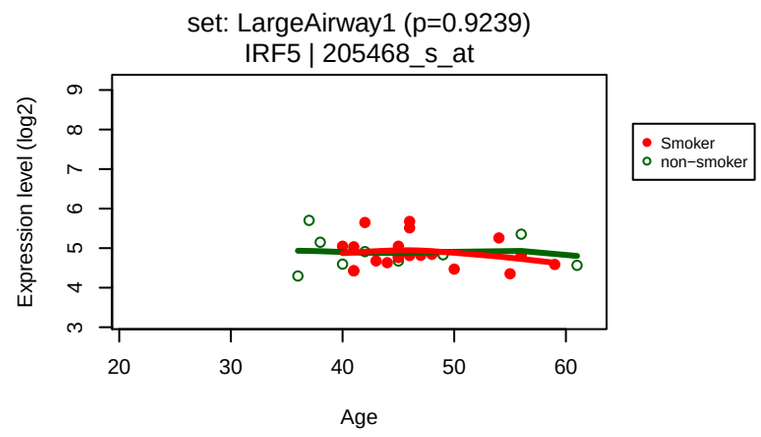

Gene expression of smokers and nonsmokers
gene: IRF5 | 205468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.364
p=0.9463
p=0.6461
p=0.9239
p=0.3459
p=0.3196
p=0.4377
p=0.0537
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

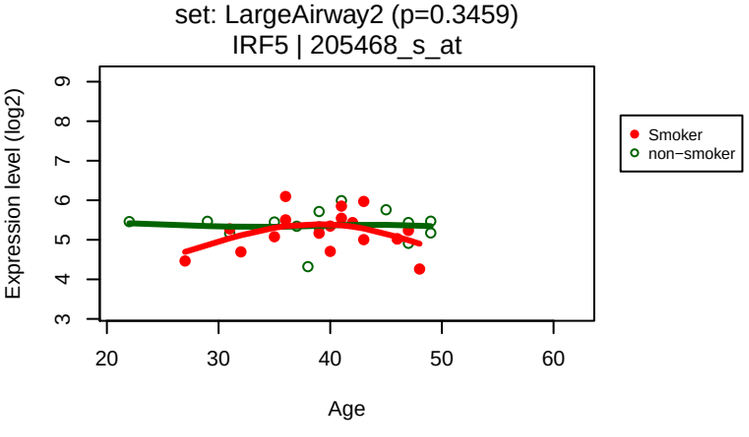

Gene expression ratio (smokers/nonsmokers)
gene: IRF5 | 205468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.364
p=0.9463
p=0.6461
p=0.9239
p=0.3459
p=0.3196
p=0.4377
p=0.0537
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

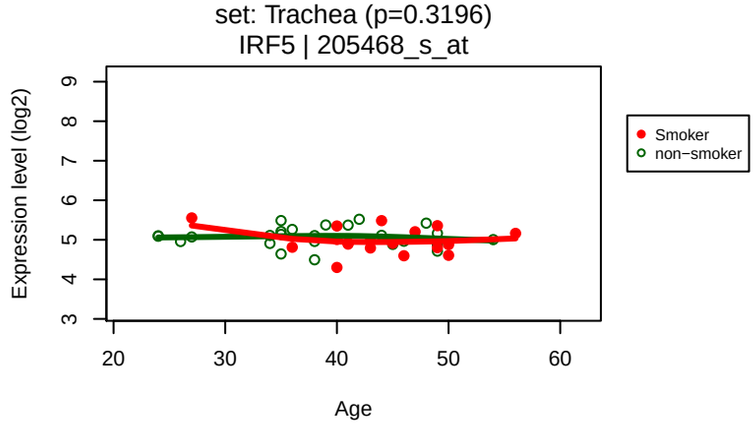

Gene expression ratio in Males
gene: IRF5 | 205468_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4493
p=0.5168
p=0.677
p=0.0134
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRF5 | 205468_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1955
p=0.0149
p=0.0411
p=0.6584
SmallAirway2
LargeAirway2
Trachea
Macrophage2

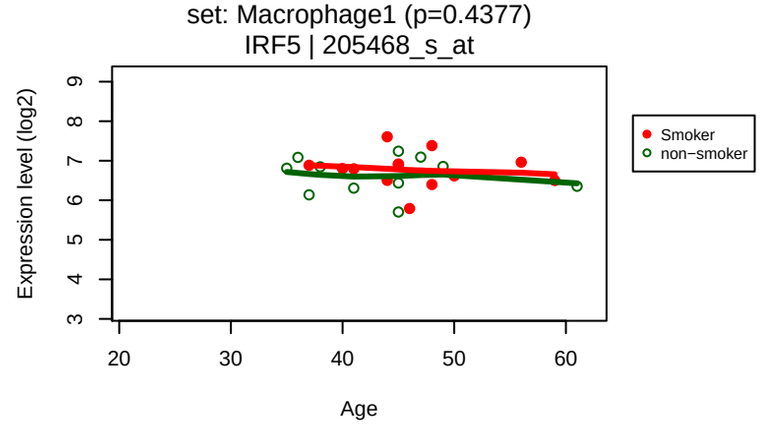

set: SmallAirway1 (p=0.364)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker

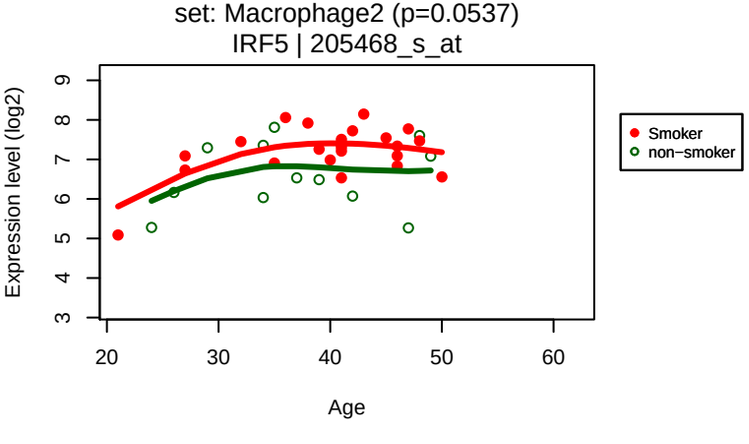

set: SmallAirway2 (p=0.9463)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker

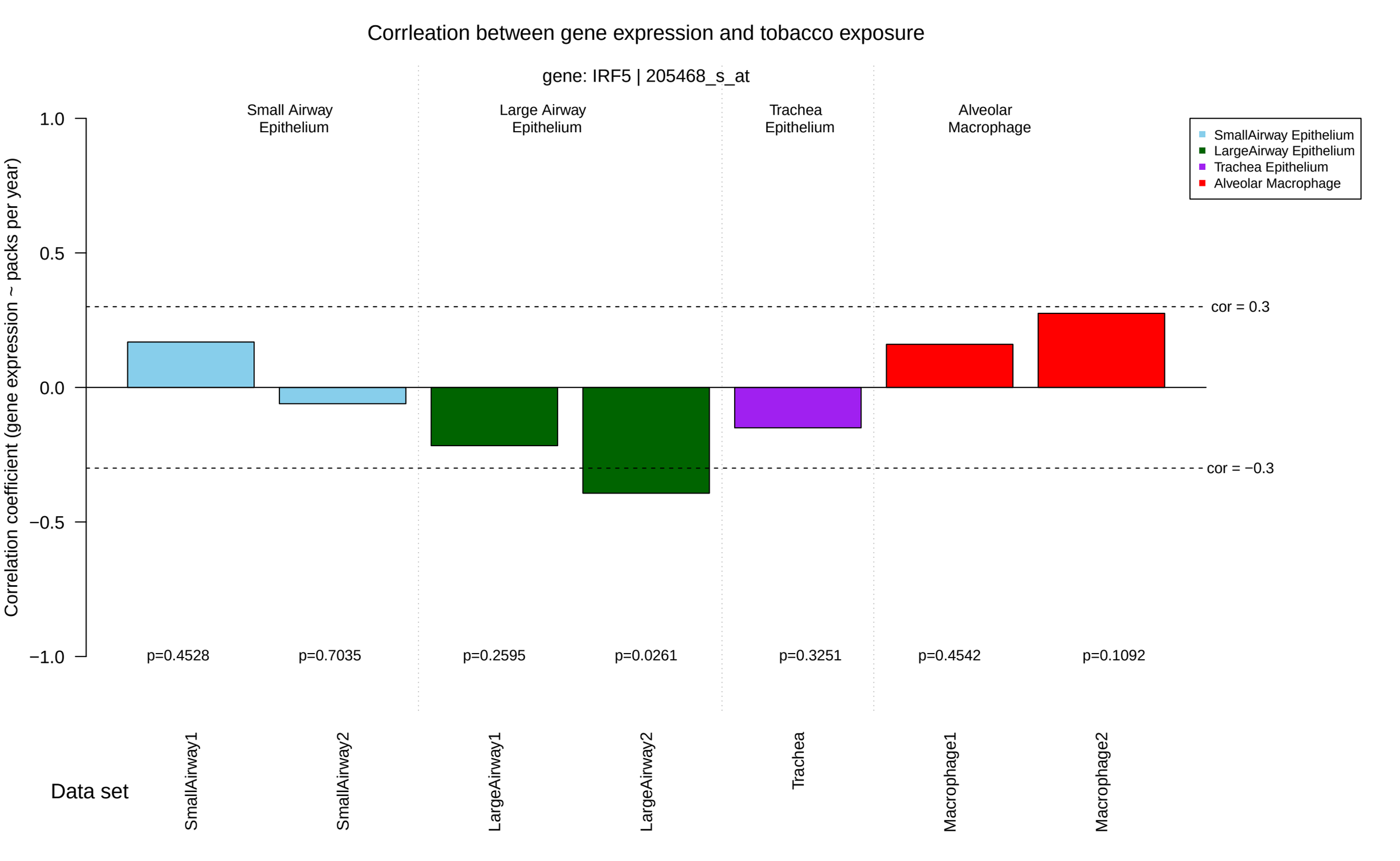

set: LargeAirway1 (p=0.9239)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3459)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3196)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4377)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0537)
IRF5 | 205468_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRF5 | 205468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4528
p=0.7035
p=0.2595
p=0.0261
p=0.3251
p=0.4542
p=0.1092
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2